

Inference rules : ways to get new expressions from known expressions.

$$\frac{P_1 \quad P_2 \quad \vdots \quad P_k}{Q}$$

$P_1, P_2, \dots, P_k \text{ infer } Q$

if $P_1, P_2, \dots, P_k$ are T then Q is T

$P_1 \wedge P_2 \wedge \dots \wedge P_k \Rightarrow Q$ is a tautology

$$P_1, P_2, \dots, P_k \vdash Q$$

↑
infer

Rule	tautology	name
$\frac{P}{P \vee q}$	$P \Rightarrow P \vee q$	addition
$\frac{P \wedge q}{P}$	$P \wedge q \Rightarrow P$	simplification
$\frac{P \quad P \Rightarrow q}{q}$ $\sim q$	$P \wedge (P \Rightarrow q) \Rightarrow q$	Modus Ponens
$\frac{P \Rightarrow q}{\sim p}$	$\sim q \wedge (P \Rightarrow q) \Rightarrow \sim P$	Modus Tollens
$\frac{P \Rightarrow q \quad q \Rightarrow r}{P \Rightarrow r}$	$(P \Rightarrow q) \wedge (q \Rightarrow r) \Rightarrow (P \Rightarrow r)$	Hypothetical Syllogism
$\frac{P \quad q}{P \wedge q}$	$P \wedge q \Rightarrow (P \wedge q)$	conjunction

example: with these hypotheses

H_1 - I wake up late

H_2 - If I wake up late then I miss the bus

H_3 - If I miss the bus then I walk to school

prove that I walked to school

C S

Proof: 1. w

Hyp.

2. $w \Rightarrow b$

Hyp.

M.P.

3. $b \Rightarrow s$

Hyp.

A

4. b

M.P ①+②

$A \Rightarrow B$

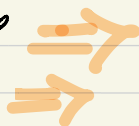
5. s

M.P ③+④

B

exmp: with these hypothesis

- it is not sunny today and it is colder than yesterday
- we will go swimming only if it is sunny
- if we do not go swimming then we will take a canoe trip
- if we take a canoe trip then we will be home by sunset



prove that we will be home by the sunset

Proof: 1. $\sim p \wedge q$ Hyp.

2. $r \Rightarrow p$ Hyp $r \vee p / r \Rightarrow p$

3. $\sim r \Rightarrow s$ Hyp.

4. $s \Rightarrow t$ Hyp.

$\frac{A \wedge B}{A}$ Simp

5. $\sim p$ simp ①

6. $\sim p \Rightarrow \sim r$ Contradiction ②

$\frac{A}{A \Rightarrow B}$ M.P.
 B

7. $\sim r$ M.P. ⑤ + ⑥

8. s M.P. ⑦ + ③

9. t M.P. ⑧ + ④

$\sim B$
 $\frac{A \Rightarrow B}{\sim A}$ M.T.

6-7. $\sim r$ M.T ⑤ + ②

if it is not sunny then we will not go swimming $\sim p \Rightarrow \sim r$
we will go swimming only if it is sunny $r \Rightarrow p$
if it is sunny then we will go swimming
Contradiction

Constructive Dilemma

$$P \Rightarrow Q$$

$$r \Rightarrow s$$

$$r \vee p$$

$$\hline s \vee q$$

Proof: $\checkmark 1. p \Rightarrow r$

$$\checkmark 2. r \Rightarrow s$$

$$\checkmark 3. r \vee p$$

$$\checkmark 4. \sim \sim r \vee p$$

$$5. \sim r \vee s$$

$$\checkmark 6. \sim r \Rightarrow r$$

$$\checkmark 7. \sim r \Rightarrow r$$

$$\checkmark 8. \sim s \Rightarrow \sim r$$

$$\checkmark 9. \sim s \Rightarrow r$$

$$\checkmark 10. \sim \sim s \vee q$$

$$\checkmark 11. s \vee q$$

Hyp

Hyp

Hyp

Double neg. (3)

Imp (2)

Imp (4)

H.S. (6) + (1)

Contrapositive (2)

H.S. (7) + (8)

Imp (9)

Double neg. (10)

$$\text{Imp } A \Rightarrow B \equiv \sim A \vee B$$

$$\text{D.N. } A \equiv \sim \sim A$$

$$\text{Cont. } A \Rightarrow B \equiv \sim B \Rightarrow \sim A$$

We never used 5.

So we can remove

$$\text{Q. } \frac{P \Rightarrow r}{p \vee s \Rightarrow q \vee s}$$

use constructive dilemma